

Toxicology

Studying how substances harm living organisms

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Toxicology studies the harmful effects of chemicals, drugs, and other substances on living organisms, determining safe exposure limits and understanding mechanisms of harm.

● **DEFINITION**

The scientific study of the adverse effects of chemical, physical, or biological agents on living organisms and the environment.

● **WHY IT MATTERS**

Toxicology determines safe exposure levels for chemicals in food, medicine, air, and consumer products, directly protecting public health.

● **WORKING PRINCIPLE**

INPUT / PROBLEM

CORE PROCESS

OUTCOME / APPLICATION

Toxicologists expose test systems to varying concentrations of a substance, measure biological effects, and use this dose-response data to establish safe exposure limits for humans and the environment.

● **QUICK FACTS**

- ▶ The saying 'the dose makes the poison' comes from 16th-century toxicologist Paracelsus
- ▶ Even water can be toxic in large enough doses

● **KEY TERMS**

Dose-response

LD50

Toxicokinetics

Risk assessment

● **CORE CONCEPTS**

- ▶ Dose-response relationship
- ▶ Acute vs. chronic toxicity
- ▶ Toxicokinetics
- ▶ Risk assessment

● **APPLICATIONS**

- ▶ Chemical safety testing
- ▶ Drug safety evaluation
- ▶ Environmental contaminant risk assessment
- ▶ Occupational exposure limit setting

● **REAL-LIFE EXAMPLES**

- ▶ Safety limits on food additives
- ▶ Workplace chemical exposure regulations
- ▶ Pesticide residue limits on produce

● **ADVANTAGES**

- ▶ Directly protects public and environmental health
- ▶ Guides safe regulatory limits for chemicals
- ▶ Essential for new drug and product approval

● **LIMITATIONS**

- ▶ Animal test results don't always predict human effects precisely
- ▶ Long-term, low-dose effects can be difficult to study
- ▶ New chemicals require years of testing

CAREER OPPORTUNITIES

Toxicologist, Regulatory scientist, Occupational health specialist, Environmental risk assessor

INDUSTRY APPLICATIONS

Pharmaceuticals, Chemical manufacturing, Regulatory agencies, Environmental consulting

RESEARCH AREAS

Computational toxicology, Nanotoxicology, Endocrine disruptor research

FUTURE SCOPE

Growing use of computational and cell-based testing methods is reducing reliance on animal testing while improving prediction accuracy.

● **CURRENT TRENDS**

Rising adoption of 'toxicology in the 21st century' approaches using human cell-based and computational models instead of animal testing.

● SUMMARY

Toxicology determines how substances harm living organisms, establishing the safe exposure limits that protect public health.